

L02 – Distribution Network Power Flow

Foundations Lecture Series

Course Instructor

Microgrid 3-phase Security AI Project

Foundations Lecture 2

Keywords: radial topology, R/X ratio, backward/forward sweep, DER, voltage rise

Lecture roadmap

- ① Why distribution differs from transmission
- ② Radial topology and the R/X problem
- ③ Backward/Forward Sweep (BFS) algorithm
- ④ DistFlow and linear voltage-drop approximations
- ⑤ DER impact: reverse flow, voltage rise, hosting capacity
- ⑥ Toward unbalanced analysis (preview of L04)

Prerequisite

L01 – you should be comfortable with Y_{bus} , NR, and `pp.runpp`.

Transmission vs distribution: a side-by-side

TODO: *Build a comparison table covering: voltage level, topology (meshed vs radial), R/X ratio, load composition, three-phase symmetry, DER penetration. Highlight that distribution breaks several assumptions L01 relied on.*

The R/X problem in one slide

TODO: *Explain that high R/X ratio (typical for LV/MV feeders) makes the standard decoupled approximation invalid. Show the simplified voltage-drop formula and note that the P -term cannot be ignored. Set up motivation for BFS.*

Backward/Forward Sweep at a glance

TODO: *Algorithm sketch: traverse radial tree from leaves to root summing currents, then from root back to leaves updating voltages. Emphasize that it exploits the radial structure and avoids the Jacobian altogether.*

When BFS beats Newton-Raphson

TODO: *Convergence comparison on radial vs lightly-meshed networks. Note that BFS is sometimes the only solver that converges on very long LV feeders. pandapower exposes BFSW via `algorithm="bfsw"`.*

TODO: *Introduce the linearized voltage-drop formula and the DistFlow / LinDistFlow framework. Explain when the linearization is accurate and when it lies (high P injection, high R/X).*

Reverse power flow and voltage rise

TODO: *Show how a high-PV bus can have $V > 1.05$ pu (voltage rise) and how PV at the end of a feeder can cause reverse flow into the substation. Explain hosting capacity in one sentence.*

The microgrid extension

TODO: Highlight grid-connected vs islanded modes, PCC role, DER + BESS as controllable resources.
Forward-reference Week 1 kickoff material.

Why distribution is rarely balanced in practice

TODO: *List sources of asymmetry: single-phase loads, single-phase rooftop PV, EV chargers, asymmetric line geometry. Conclude that L03's balanced assumption is convenient but not realistic for LV microgrids. Bridge to L04.*

Homework

TODO: *Suggest: implement BFSW by hand on a 5-bus radial feeder; vary R/X ratio and compare to NR; analyze a feeder with one high-PV bus and observe voltage rise.*

Exit checklist

TODO: *Fill once content frames are finalized.*

TODO: *Baran & Wu (DistFlow), Cespedes (1990 BFS paper), Kersting textbook for distribution analysis, pandapower BFSW documentation.*